St. Francis Institute of Technology
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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 10: Orchestration using Kubernetes

Aim: To study and implement container orchestration using Kubernetes on AWS/ Azure/ Google cloud platform (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –
- fundamental concepts of containerization technologies like kubernetes
- how to provision and manage container on AWS cloud with EKS Elastic Kubernetes Services.
- how to use eksctl, docker and kubectl command line
- how to create and use EKS Cluster

Lab objective mapped: ITDO8024.6 : To design and develop containerized solutions.

Prerequisite: An AWS account

Requirements:
- Must have Hyper V capabilities on the system
- Minikube installed
- Idea of system administration and linux commands
- Understands the basic concepts of containerization technology
- Cloud Login, Desktop, Browser, Internet

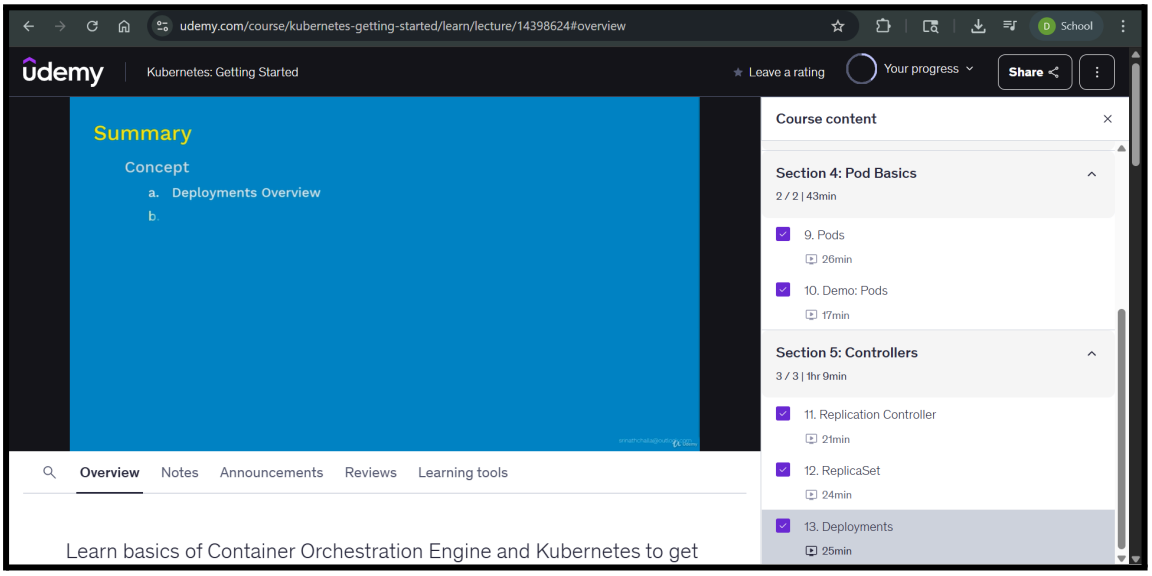
Pre-Experiment Theory: (To be written and drawn in hand)
1. **Amazon EKS**
2. **Architecture of Kubernetes**

Procedure:
Follow given aws documentation link to complete Expt 10
https://docs.aws.amazon.com/eks/latest/userguide/getting-started-console.html

Post-Experiments Exercise:

Complete Certification Course:

I. https://www.udemy.com/course/kubernetes-getting-started/
(Attach some proof of completion)



Extended Theory: (to be written in hand)

Kubernetes with AWS Elastic Kubernetes Services (EKS) (Udemy Video):

I. https://www.udemy.com/course/master-kubernetes-with-aws-elastic-kubernetes-services-eks/
Watch Preview videos
1. What is Kubernates and EKS
2. Setting up environment for Kubernates
3.

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Sr. No	**Item**	**Answer**
1.	For 2 Amazon EKS cluster that you create in Asia Pacific (Mumbai) how much does that charge	
2.	kubectl- Which tool is this? its use	

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1] [Online]https://www.bluematador.com
[2] [Online] https://docs.aws.amazon.com/eks/latest/userguide/what-is-eks.html
[3] [Online] https://www.eksworkshop.com/docs/introduction
[4] [Online] https://aws.amazon.com/getting-started/hands-on/deploy-kubernetes-app-amazon-eks/